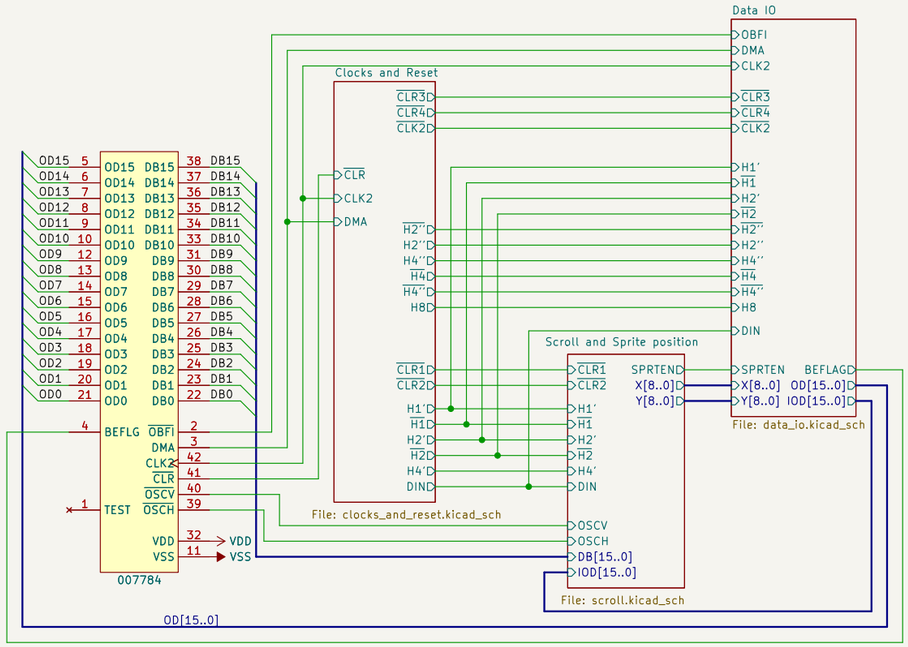
Words are parsed and written to sprite RAM in the following way:

Word 0 -> Sets/Clears BEFLAG according to bit 15.
Word 2 -> Word 3
Word 3 -> Word 0
Word 4 & 5 – XSCROLL -> Word 1
Word 6 & 7 – YSCROLL -> Word 2

The 007784 does the following things:

- Sets BEFLAG high if bit OD15 is set on every eight words in. Otherwise BEFLAG goes low. DMA does not need to be active.
- All other words in need DMA to be active to be parsed.
- Calculates the final vertical and horizontal positions of the sprite by using the sprite position in the unparsed object table and the horizontal and vertical scroll registers. It is written to a new parsed object table in RAM.
- The Sprite Enable (SPRTEN) bit is only set if the position is within the allowed range.
- When OBFI (OBJBUFINIT) in is high, only zeros are written to data out. That is to reset the object RAM table.



Ulf Skutnabba, twitter: @skutis77

Sheet: /
File: 007784.kicad_sch

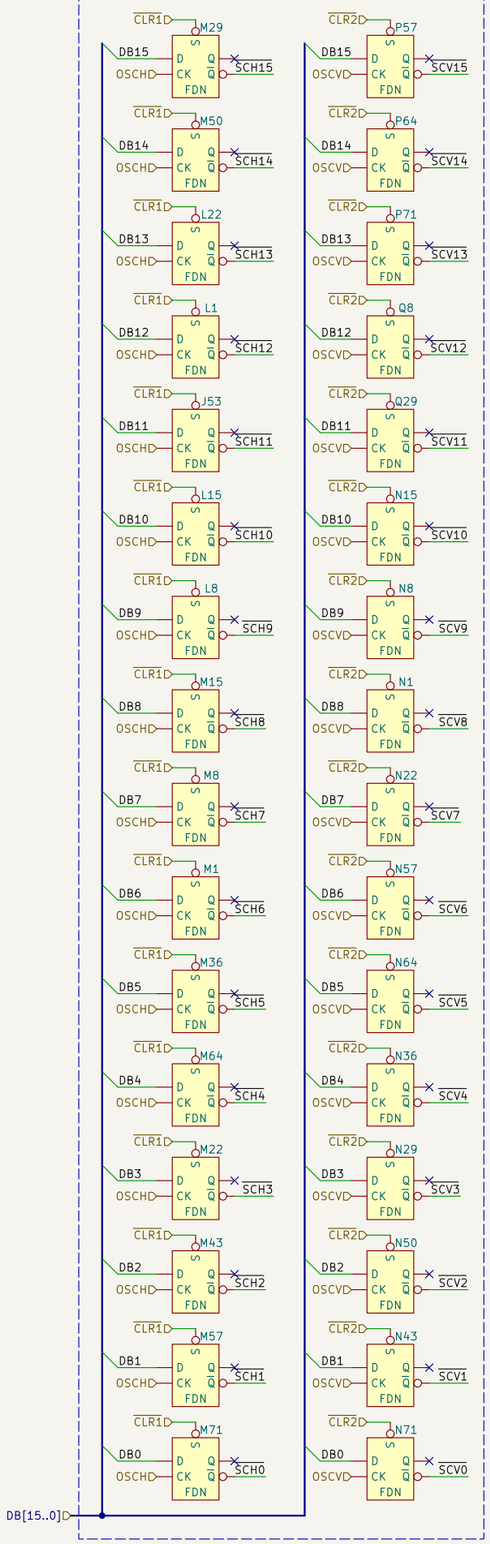
Title: Konami 007784

Size: A3	Date: 2023-10-23	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/4

Scroll Registers

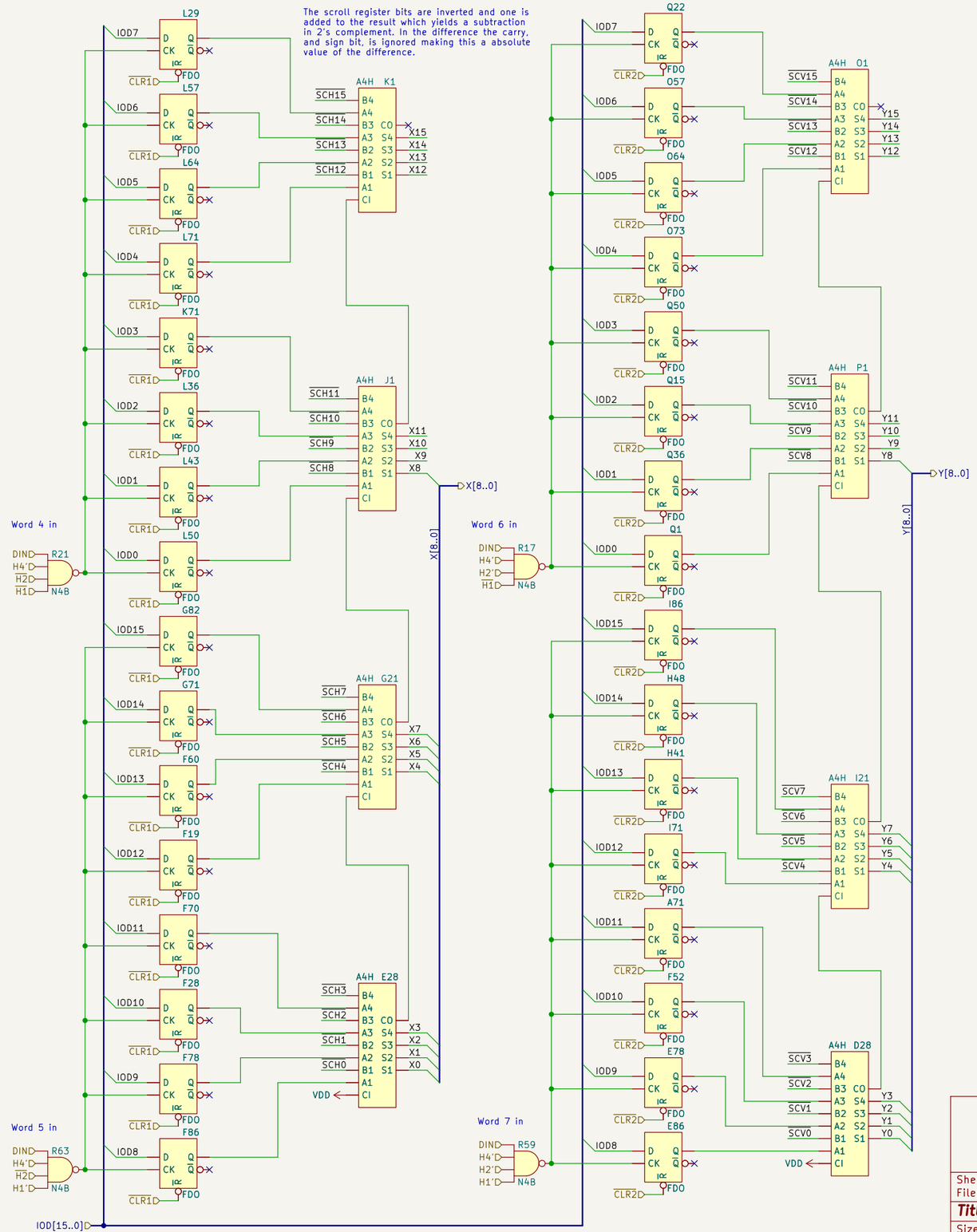
Horizontal

Vertical



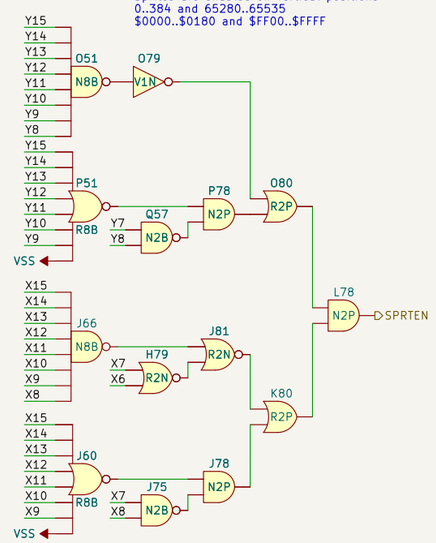
Horizontal and Vertical Positions

The scroll register bits are inverted and one is added to the result which yields a subtraction in 2's complement. In the difference the carry, and sign bit, is ignored making this a absolute value of the difference.



Sprite Enable

Sprites are enabled on vertical positions
0..384 and 65280..65535
$0000..$0180 and $FF00..$FFFF



Sprites enabled between horizontal positions
0..384 and ...

Sheet: /Scroll and Sprite position/
File: scroll.kicad_sch

Title: Konami 007784

Size: A3	Date: 2023-10-23	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/4

Word 0 in

Word 3, bit 15, out is SPRTEN (Sprite Enable)

Word 0 out

Word 1 out

Word 2 out

Word 3 out

Word 2 in

Word 3 in

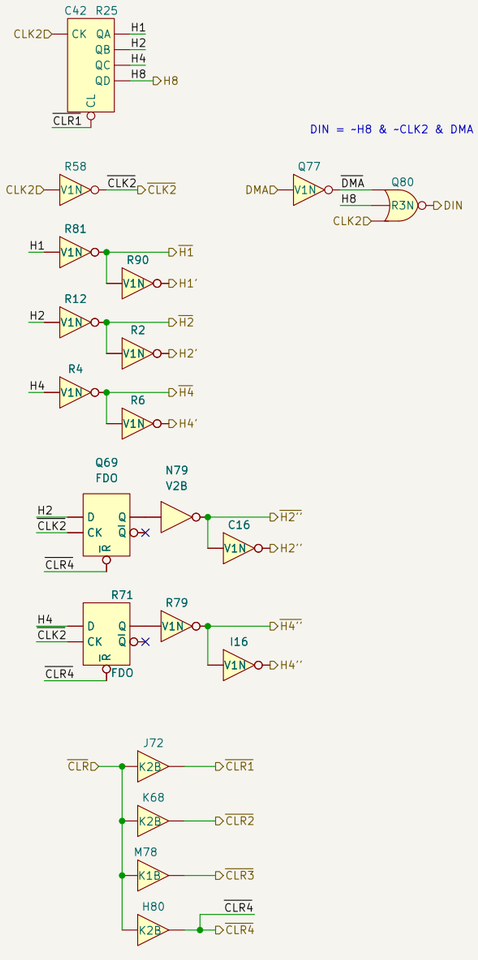
OD[15..0]

IOD[15..0]

X[8..0]

Y[8..0]

Sheet: /Data IO/	
File: data_io.kicad_sch	
Title: Konami 007784	
Size: A3	Date: 2023-10-23
KiCad E.D.A. 10.0.4	Id: 3/4



Sheet: /Clocks and Reset/		
File: clocks_and_reset.kicad_sch		
Title: Konami 007784		
Size: A3	Date: 2023-10-23	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/4